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MGR INŻ. OLEKSANDR YEROSHKIN, 24.10.2023

Mobile Applications for IOS

Presentation of basic elements of Swift by examples. Core libraries

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PROGRAMME CONTENT		
Lecture		Number of hours
Lec 1	Introduction. Review of selected mobile applications. Introduction to Apple devices and Mac OS.	2
Lec 2-3	UI and UX. Interactions. Touch screens. Material design. Human Interface guidelines. Voice communication. Siri.	4
Lec 4	Tools for multimedia processing. Gathering multimedia data. Copyright.	2
Lec 5	Implementation rules in Xcode. The structure of application written in Swift.	2
Lec 6-7	Presentation of basic elements of Swift by examples. Core libraries.	4
Lec 8	Apple Human Interface Guideline by examples.	2
Lec 9	Libraries and frameworks supporting creation of multimedia applications. Short characteristics of Kotlin language.	2
Lec 10	Applications of augmented reality. Code analysis of application with augmented reality mechanisms.	2
Lec 11	Games review. Connections between games and the progress of algorithms and programming languages.	2
Lec 12	Multimedia data compression. Compression formats. Video and audio streaming.	2
Lec 13	Multimedia in mobile systems. Cameras. Recommended frameworks.	2
Lec 14	Interaction mechanisms – review. New technologies and multimedia devices.	2
Lec 15	Summary. Perspectives of multimedia techniques.	2
	Total hours	30



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Attendance check



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Overview

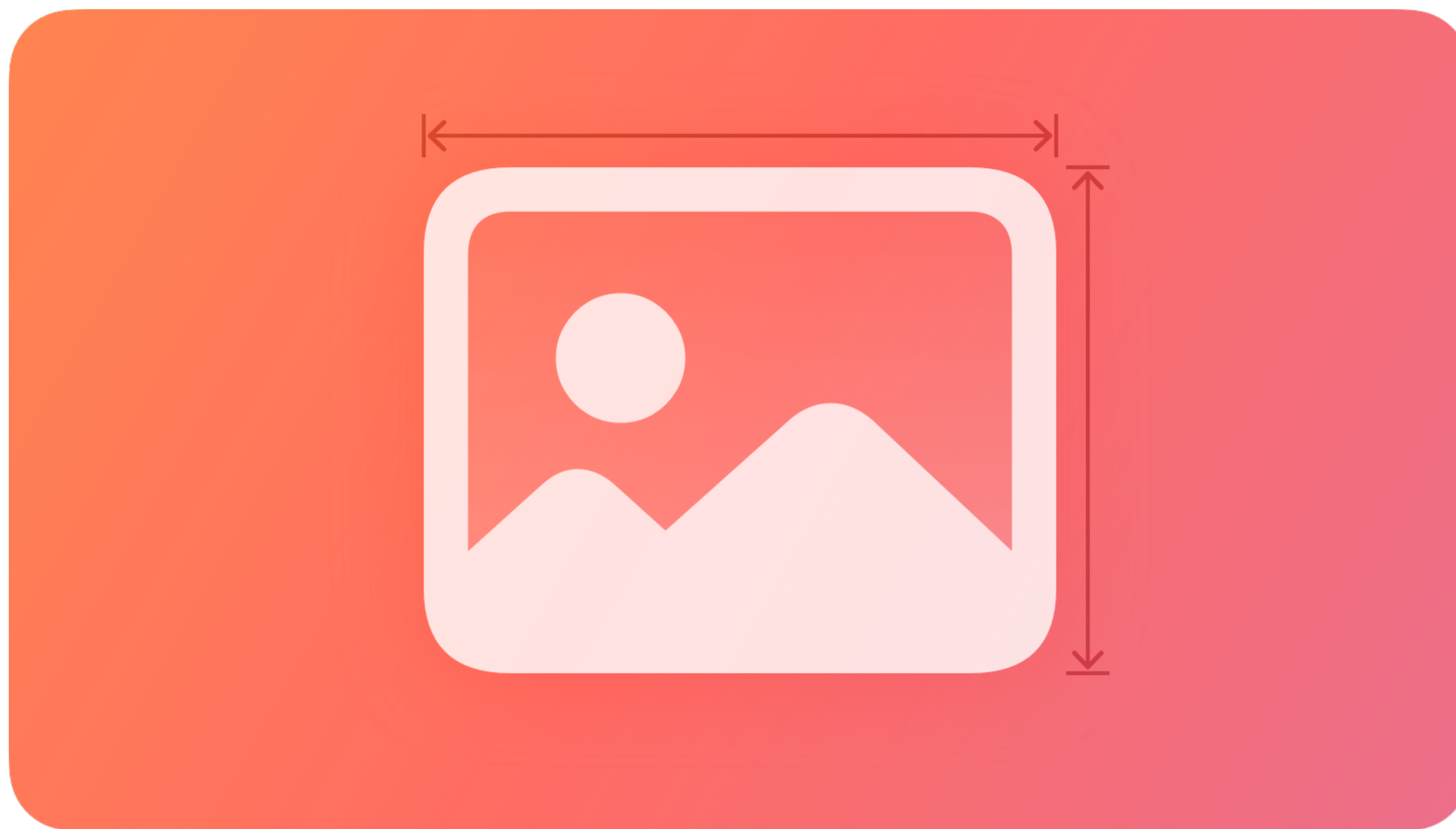
Image views, Charts, API, JSON, MVVM



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Image views





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Image views

```
Image("Landscape_4")  
    .resizable()  
    .aspectRatio(contentMode: .fit)  
Text("Water wheel")
```



Water wheel



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SwiftUI CachedAsyncImage

🔗 SwiftUI CachedAsyncImage 📱

`CachedAsyncImage` is `AsyncImage`, but with cache capabilities.

Usage

`CachedAsyncImage` has the exact same API and behavior as `AsyncImage`, so you just have to change this:

```
AsyncImage(url: logoURL)
```



to this:

```
CachedAsyncImage(url: logoURL)
```



In addition to `AsyncImage` initializers, you have the possibilities to specify the cache you want to use (by default `URLCache.shared` is used), and to use `URLRequest` instead of `URL`:

```
CachedAsyncImage(urlRequest: logoURLRequest, urlCache: .imageCache)
```



```
// URLCache+imageCache.swift

extension URLCache {

    static let imageCache = URLCache(memoryCapacity: 512_000_000, diskCapacity: 10_000_000_000)
}
```



Remember when setting the cache the response (in this case our image) must be no larger than about 5% of the disk cache (See [this discussion](#)).

Installation

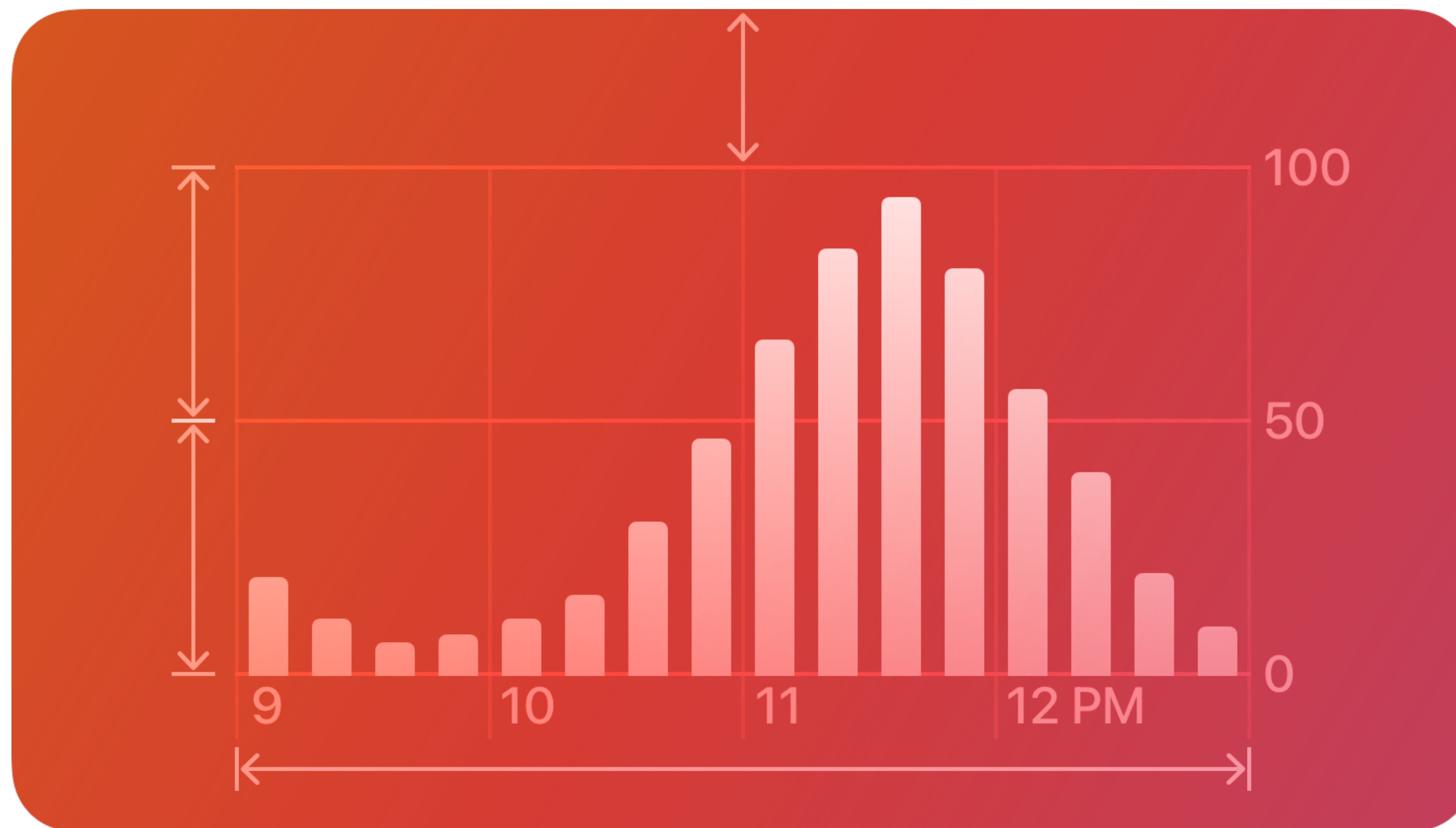
1. In Xcode, open your project and navigate to **File** → **Swift Packages** → **Add Package Dependency...**
2. Paste the repository URL (<https://github.com/lorenzofiamingo/swiftui-cached-async-image>) and click **Next**.
3. Click **Finish**.



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Charts

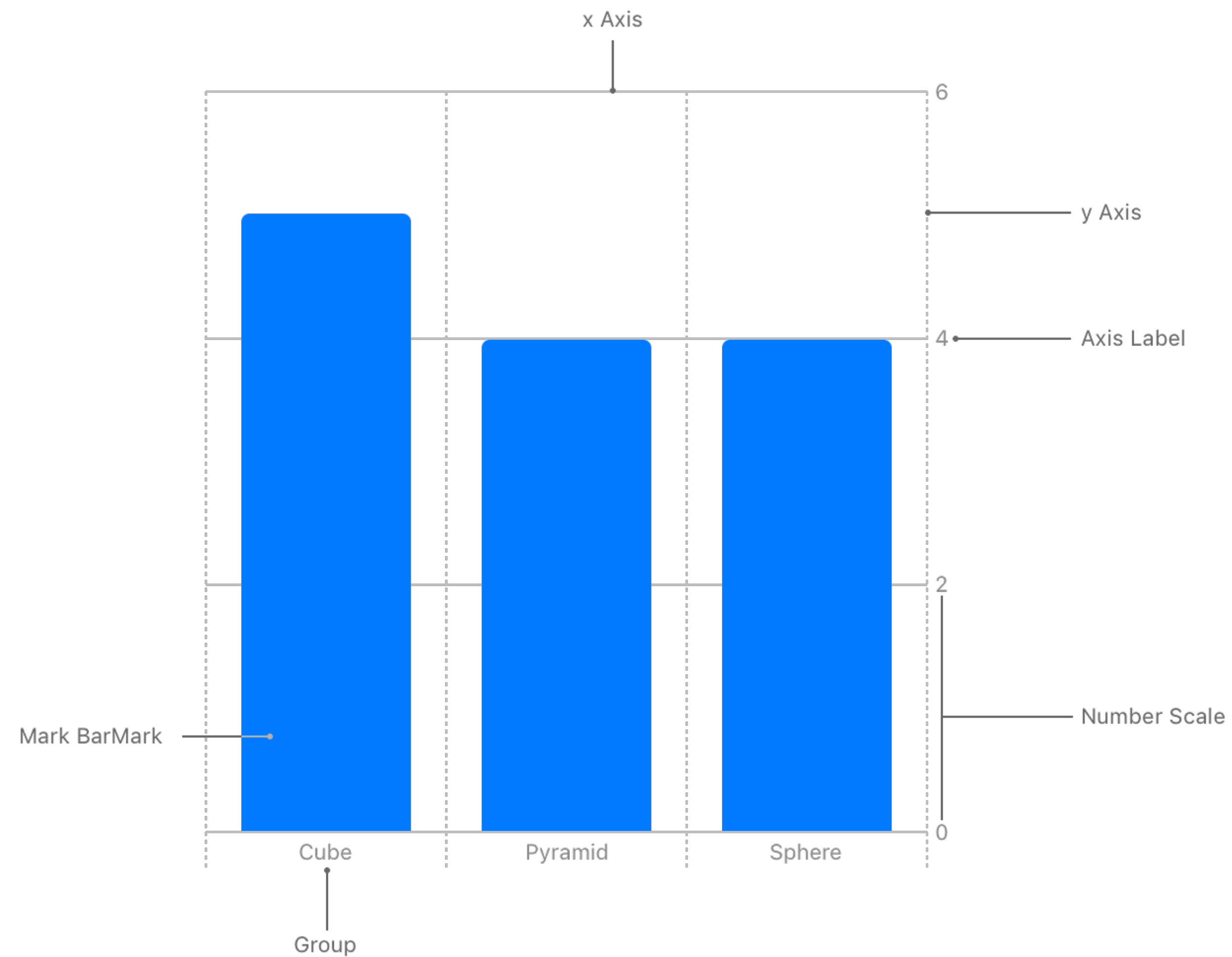




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Charts





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DEMO

Bad MVVM vs. Good MVVM



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DEMO

Image views, CachedAsyncImage, API, JSON, MVVM



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Summary and Questions



“What we know is a drop, what we
don't know is an ocean.”

Isaac Newton
